Sampling distribution of a proportion

A1S2 PRE-ASSESSMENT STUDY

1 RECOGNISE IT

- Each observation is success/failure.
- The random statistic is p=X/n.
- Population proportion π is known for the probability model.

2 DECISION / METHOD

1. Check nπ > 5 AND n(1-π) > 5.
2. Centre at π.
3. Use theoretical SE with π.
4. Standardise the p boundary and use normal areas.

3 FORMULA

p=X/n
E(p)=π
SE(p)=sqrt(π*(1-π)/n)
z=(p-π)/SE(p)

4 FORMULA SHEET VS WHAT I KNOW

Gives: p estimates π; theoretical SE uses π; Z model.
Still know: approximation conditions and p versus π roles.

5 EXCEL ROUTE

Direct left tail:
=NORM.DIST(p_boundary,pi,SQRT(pi*(1-pi)/n),TRUE)
Use 1-minus/right or upper-minus-lower.

6 TINY WORKED EXAMPLE

π=.40, n=100. For P(p<.50), SE=.0490 and z=2.041; probability is about .9794.

7 TOP EXAM TRAPS

- Weakening the strict sourced course inequality.
- Substituting observed p into this theoretical probability SE.
- Treating p as a count.
- Forgetting that π is fixed and p is random.

8 SAY ALOUD

For sampling probabilities, p is random, π is the centre, and theoretical SE uses π.

9 EXIT CHECK

Which symbol belongs in SE(p) for a sampling-distribution probability: p or π?

ONE-GLANCE VISUAL ROUTE

